

MUHAMMAD HISYAM

Full Stack & AI Software Engineer

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Portfolio: mhisyam.com · Location: Bekasi, Indonesia

PROFESSIONAL SUMMARY

Full Stack & AI Software Engineer with extensive expertise in building desktop cross-platform applications, high-performance web systems, medical deep learning pipelines, and agentic AI integrations. Skilled across TypeScript, Rust, Python, Golang, Java, and Supabase/PostgreSQL. Creator of 26+ open-source and commercial-grade projects including RetailHub (Tauri + Supabase POS platform), Pneumonia U-Net Segmentation, and Gemini MCP Client.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, Golang, Java, PHP, PL/pgSQL, HTML5, CSS3, LaTeX
Desktop & Frontend: Tauri, Astro 5, React.js, SolidJS, SvelteKit, Next.js, Vue.js, Tailwind CSS, TanStack Router
Backend & Cloud: Supabase, Node.js, Express.js, Go (Gin/Fiber), Java Spring Boot, Laravel, REST APIs, WebSockets
AI & Data Engineering: PyTorch, U-Net, OpenCV, Model Context Protocol (MCP), Gemini API
Databases & DevOps: PostgreSQL, Supabase RLS, MongoDB, Redis, Docker, Git, GitHub Actions, Vite, Bun, Biome, Linux CLI

FEATURED PROJECTS

- RetailHub – Retail Management & POS Platform** (Tauri + Rust + Supabase) [\[Repository Link\]](#)
- Engineered a desktop and cross-platform Retail Point-of-Sale (POS) and inventory management platform using Tauri (Rust backend runtime), TypeScript, Vite, and TanStack Router.
 - Configured Supabase (PostgreSQL) data architecture with Row Level Security (RLS) policies, PL/pgSQL database functions, and real-time database subscription sync.
- Developer Portfolio & Interactive Hub** (Astro 5 + SolidJS) [\[Repository Link\]](#)
- Engineered a high-speed developer portfolio using Astro 5, SolidJS, and Tailwind CSS 4.
 - Integrated an embedded Dev Suite featuring a retro Unix terminal shell (`hisyam-cli`), REST API response demo client, audio music player, and 8 mini-games arcade.
- Pneumonia Segmentation U-Net Model** (Python + Deep Learning) [\[Repository Link\]](#)
- Developed a deep learning semantic segmentation pipeline utilizing PyTorch and U-Net architecture for chest X-ray pneumonia detection with evaluation routines.
- Gemini Model Context Protocol (MCP) Client** (TypeScript) [\[Repository Link\]](#)
- Built an extensible MCP client enabling agentic AI workflows and tool-use integrations with Google Gemini models.
- Real-Time Energy Monitoring System** (TypeScript + IoT) [\[Repository Link\]](#)
- Built a real-time energy telemetry analytics dashboard processing high-frequency sensor streams with dynamic live charts.
- Golang & SvelteKit Real-Time Chat App** (Go + WebSockets) [\[Repository Link\]](#)
- Implemented a concurrent real-time chat application with a Golang WebSocket backend server and SvelteKit reactive frontend.
- Secure JWT Authentication Service** (Java) [\[Repository Link\]](#)
- Designed a robust Java authentication microservice supporting access tokens, sliding session refresh tokens, and instant token version invalidation.

EDUCATION & COMMUNITY

- Universitas Gunadarma** *Bekasi, Indonesia*
Bachelor of Computer Science (Informatika) — Fresh Graduate
- Focus on Software Engineering, Web Systems, and Applied Artificial Intelligence.
 - Maintainer of 26+ open-source GitHub repositories spanning AI, web apps, backend APIs, and developer utilities.